center of the bowl as in Figure 9.1b; they have zero energy when their value is zero. That makes good, intuitive sense since we associate emptying a region of space with setting everything, including field values, to zero.

But for a Higgs field, things work differently. Just as a frog can reach the central plateau in Figure 9.1c and be *zero* distance from the worm pile only if it has enough energy to jump up from the surrounding valley, a Higgs field can reach the bowl's center, and have *value zero,* only if it too embodies enough energy to surmount the bowl's central bump. If, to the contrary, the frog has little or no energy, it will slide to the valley in Figure 9.1d—a *nonzero* distance from the worm pile. Similarly, a Higgs field with little or no energy will also slide to the bowl's valley—a nonzero distance from the bowl's center—and hence it will have a *nonzero* value.

To force a Higgs field to have a value of zero—the value that would seem to be the closest you can come to completely removing the field from the region, the value that would seem to be the closest you can come to a state of nothingness—you would have to *raise* its energy and, energetically speaking, the region of space would not be as empty as it possibly could. Even though it sounds contradictory, removing the Higgs field— reducing its value to zero, that is—is tantamount to adding energy to the region. As a rough analogy, think of one of those fancy noise reduction headphones that produce sound waves to cancel those coming from the environment that would otherwise impinge on your eardrums. If the headphones work perfectly, you hear silence when they produce their sounds, but you hear the ambient noise if you shut them off. Researchers have come to believe that just as you hear *less* when the headphones are suffused with the sounds they are programmed to produce, so cold, empty space harbors as little energy as it possibly can —it is as empty as it can be—when it is suffused with an ocean of Higgs field. Researchers refer to the emptiest space can be as the *vacuum,* and so we learn that the vacuum may actually be permeated by a uniform Higgs field.

The process of a Higgs field's assuming a nonzero value throughout space—forming a Higgs ocean—is called *spontaneous symmetry*